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MODULE -1

Introduction to Cyber Law

How Computers Evolved?

Computers in the form of personal desktop computers, laptops and tablets have become such an important part of everyday living that it can be difficult to remember a time when they did not exist. In reality, computers as they are known and used today are still relatively new. Although computers have technically been in use since the abacus approximately 5000 years ago, it is modern computers that have had the greatest and most profound effect on society. The first full-sized digital computer in history was developed in 1944. Called the Mark I, this computer was used only for calculations and weighed five tons. Despite its size and limited ability it was the first of many that would start off generations of computer development and growth.

First Generation Computers

First generation computers bore little resemblance to computers of today, either in appearance or performance. The first generation of computers took place from 1940 to 1956 and was extremely large in size. The inner workings of the computers at that time were unsophisticated. These early machines required magnetic drums for memory and vacuum tubes that worked as switches and amplifiers. It was the vacuum tubes that were mainly responsible for the large size of the machines and the massive amounts of heat that they released. These computers produced so much heat that they regularly

overheated despite large cooling units. First generation computers also used a very basic programming language that is referred to as machine language.

Second Generation Computers

The second generation (from 1956 to 1963) of computers managed to do away with vacuum tubes in lieu of transistors. This allowed them to use less electricity and generate less heat. Second generation computers were also significantly faster than their predecessors. Another significant change was in the size of the computers, which were smaller. Transistor computers also developed core memory which they used alongside magnetic storage.

Third Generation Computers

From 1964 to 1971 computers went through a significant change in terms of speed, courtesy of integrated circuits. Integrated circuits, or semiconductor chips, were large numbers of miniature transistors packed on silicon chips. This not only increased the speed of computers but also made them smaller, more powerful, and less expensive. In addition, instead of the punch cards and the printouts of

previous systems, keyboards and monitors were now allowing people to interact with computing machines.

Fourth Generation Computers

The changes with the greatest impact occurred in the years from 1971 to 2010. During this time technology developed to a point where manufacturers could place millions of transistors on a single circuit chip. This was called monolithic integrated circuit technology. It also heralded the invention of the Intel 4004 chip which was the first microprocessor to become commercially available in 1971. This invention led to the dawn of the personal computer industry. By the mid-70s, personal computers such as the Altair 8800 became available to the public in the form of kits and required assembly. By the late 70s and early 80s assembled personal computers for home use, such as the Commodore Pet, Apple II and the first IBM computer, were making their way onto the market. Personal computers and their ability to create networks eventually would lead to the Internet in the early 1990s. The fourth generation of computers also saw the creation of even smaller computers including laptops and hand-held devices. Graphical user interface, or GUI, was also invented during this time. Computer memory and storage also went

through major improvements, with an increase in storage capacity and speed.

The Fifth Generation of Computers

In the future, computer users can expect even faster and more advanced computer technology. Computers continue to develop into advanced forms of technology. Fifth generation computing has yet to be truly defined, as there are numerous paths that technology is taking toward the future of computer development. For instance, research is ongoing in the fields of nanotechnology, artificial intelligence, as well as quantum computation.

Emergence of cyber space

cyberspace, [amorphous](#), supposedly “virtual” world created by links between [computers](#), [Internet](#)-enabled devices, servers, routers, and other components of the Internet’s [infrastructure](#). As opposed to the Internet itself, however, cyberspace is the place produced by these links. It exists, in the perspective of some, apart from any particular nation-state.

The term *cyberspace* was first used by the American-Canadian author [William Gibson](#) in 1982 in a story published

in *Omni* magazine and then in his book [Neuromancer](#). In this science-fiction novel, Gibson described cyberspace as the creation of a [computer network](#) in a world filled with [artificially intelligent](#) beings.

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Cyber Jurisprudence

Introduction

Jurisprudence can be defined as the science and philosophy or theory of the law. Applying jurisprudence to cyber law gives rise to the legal study that concentrates on the logical structure, the meanings and uses of its concepts, and the formal terms and modes of operation of cyber law. Cyberlaw is a very recent concept and if compared with other older branches of the law, is a little structured study.

The term cyberspace was originally coined by a science fiction writer William Gibson to depict data matrices existing in a dark distant future which means the information spaces made by the technology of digital networked computer systems that ultimately connect with the mother of all networks that is the Internet. With the advent of the internet and technology, cyberspace along with a number of crimes related to the same emerged and expanded. As we enter the cyber age, the law on all fronts is struggling to keep pace with technological advances in cyberspace. While there is a prosperous discussion of the nature of cyber law and its challenges, still a

fundamental body of scholarly contributions to the discussion is lacking. The outgrowth of cyber jurisprudence around the world has promoted the emergence of newer dimensions in Law. The focus is on the practical aspect of cybercrime with the initial attempt to extend the known physical society concepts to the virtual space rather than the theory, philosophy, and science of cyberlaw generally. Hence in due course, we need to develop separate Cyber Jurisprudence to deal with future disputes.

The modern jurists have been cautious to endow with the rationale pedestal of jurisprudence to this ruling and now ascertained utmost exact definition of cyber jurisprudence as this describes the principles of legal issues, which exclusively

regulates the cyberspace and internet can be termed as cyber jurisprudence with a virtual approach.

Jurisprudential Aspects of Cyber Laws

Cyber jurisprudence gives an analysis of the land with land and no border, different from the physical world, they may be virtual from origin and nature. This covers the virtual world with virtual rules and policies, along with the virtual subject matter, virtual contracts, virtual disputes, virtual property, virtual possession, and virtual court.

The existence of an item in the context of a virtual world, such as an e-mail account or an online game, is also a form of virtual property. It emphasizes the composite idea of cyber jurisdiction, cyber court's venue in the cyberspace, and recognize uniform cyber rules and policies at the international level. Framing rules and laws to cover every aspect will be an arduous task since the cyber world has no boundaries.

However, a balance has to be maintained and laws be evolved in order to keep a check on cybercrimes. Whenever a conflict is encountered in implementing existing laws of the real space to Cyber Space, the laws of the real space have prevailed, overtime this tendency is likely to develop into a principle of "Primacy of Meta Space" and become the bedrock of Jurisprudence. However, the principle fails when two laws of the real space itself come into conflict in the Cyber Space.

Doctrinal approach

Cyber Law also called IT Law is the law regarding Information-technology including computers and the internet. It is related to legal informatics and supervises the digital circulation of information, software, information security, and e-commerce.

IT law does not consist of a separate area of law rather it encloses aspects of contract, intellectual property, privacy, and data protection laws. Intellectual property is a key element of IT law. The area of software license is controversial and still evolving in Europe and elsewhere.

According to the Ministry of Electronics and Information Technology, Government of India :

Cyber Laws yields legal recognition to electronic documents and a structure to support e-filing and e-commerce transactions and also provides a legal structure to reduce, check cyber crime

Importance of Cyber Law:

1. It covers all transactions over the internet.
2. It keeps eye on all activities over the internet.
3. It touches every action and every reaction in cyberspace.

Area of Cyber Law:

Cyber laws contain different types of purposes. Some laws create rules for how individuals and companies may use computers and the internet while some laws protect people from becoming the victims of crime through unscrupulous activities on the internet. The major areas of cyber law include:

1. *Fraud:*

Consumers depend on cyber laws to protect them from online fraud. Laws are made to prevent identity theft, credit card theft, and other financial crimes that happen online. A person who commits identity theft may face confederate or state criminal charges. They might also encounter a civil action brought by a victim. Cyber lawyers work to both defend and prosecute against allegations of fraud using the internet.

2. *Copyright:*

The internet has made copyright violations easier. In the early days of online communication, copyright violations were too easy. Both companies and individuals need lawyers to bring an action to impose copyright protections.

Copyright violation is an area of cyber law that protects the rights of individuals and companies to profit from their creative works.

3. ***Defamation:***

Several personnel uses the internet to speak their mind. When people use the internet to say things that are not true, it can cross the line into defamation. Defamation laws are civil laws that save individuals from fake public statements that can harm a business or someone's reputation. When people use the internet to make statements that violate civil laws, that is called Defamation law.

4. ***Harassment and Stalking:***

Sometimes online statements can violate criminal laws that forbid harassment and stalking. When a person makes threatening statements again and again about someone else online, there is a violation of both civil and criminal laws. Cyber lawyers both prosecute and defend people when stalking occurs using the internet and other forms of electronic communication.

5. ***Freedom of Speech:***

Freedom of speech is an important area of cyber law. Even though cyber laws forbid certain behaviors online, freedom of speech laws also allows people to speak their minds. Cyber lawyers must advise their clients on the limits of free speech including laws that prohibit obscenity. Cyber lawyers may also defend their clients when there is a debate about whether their actions consist of permissible free speech.

6. ***Trade Secrets:***

Companies doing business online often depend on cyber laws to protect their trade secrets. For example, Google and other online search engines spend lots of time developing the algorithms that produce search results. They also spend a great deal of time developing other features like maps, intelligent assistance, and flight search services to name a few. Cyber laws help these companies to take legal action as necessary to protect their trade secrets.

7. ***Contracts and Employment Law:***

Every time you click a button that says you agree to the terms and conditions of using a website, you have used cyber law. There are terms and conditions for every website that are somehow related to privacy concerns.

Consensual approach

A consensual definition allows future research to be aligned and it facilitates the interpretation and comparison of existing research. The findings suggest that the routine activity approach can be applied to the digital world

Cyber Ethics

cyber ethics is the [philosophic](#) study of [ethics](#) pertaining to computers, encompassing user behavior and what computers are programmed to do, and how this affects individuals and society. For years, various governments have enacted regulations while organizations have defined policies about cyberethics.

In the late 19th century, the invention of cameras spurred similar ethical debates as the internet does today. During a seminar of *Harvard Law Review* in 1890, Warren and Brandeis defined privacy from an ethical and moral point of view to be:

"central to dignity and individuality and boyhood. Privacy is also indispensable to a sense of autonomy — to 'a feeling that there is an area of an individual's life that is totally under his or her control, an area that is free from outside intrusion.' The deprivation of privacy can even endanger a person's health."

Over 100 years later, the internet and proliferation of private data through governments and [ecommerce](#) is an area which requires a new round of ethical debate involving a person's privacy.

Privacy can be decomposed to the limitation of others' access to an individual with "three elements of secrecy, anonymity, and

solitude." Anonymity refers to the individual's right to protection from undesired attention. Solitude refers to the lack of physical proximity of an individual to others. Secrecy refers to the protection of personalized information from being freely distributed.

Individuals surrender private information when conducting transactions and registering for services. Ethical business practice protects the privacy of their customers by securing information which may contribute to the loss of secrecy, anonymity, and solitude. Credit card information, social security numbers, phone numbers, mothers' maiden names, addresses and phone numbers freely collected and shared over the internet may lead to a loss of Privacy.

Fraud and impersonation are some of the malicious activities that occur due to the direct or indirect abuse of private information. Identity theft is rising rapidly due to the availability of private information in the internet. For instance, seven million Americans fell victim to identity theft in 2002, and nearly 12 million Americans were victims of identity theft in 2011 making it the fastest growing crime in the United States.¹ Public records search engines and databases are the main culprits contributing to the rise of cybercrime. Listed below are a few recommendations to restrict online databases from proliferating sensitive personnel information.

1. Exclude sensitive unique identifiers from database records such as social security numbers, birth dates, hometown and mothers' maiden names.
 2. Exclude phone numbers that are normally unlisted.
 3. Clear provision of a method which allows people to have their names removed from a database.
 4. Banning the reverse social security number lookup services.
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Cyber Jurisdiction

A fast-paced world, and surprisingly fitting in one's hand. The world is in the era of "internet and cyberspace", and it seems faster and better than ever. But it all comes with a price, that mankind is still in the exploration of. Just as in the real and physical world, the virtual space created by humans also sees a plethora of criminal activities on a day to day basis where the data of millions of people acts as valuable assets. It has the power to instigate a civil war or to destroy nations altogether, steal data for ransom, or even rob millions from a bank in seconds. It becomes quite a challenge to map out a conclusive set of applicable laws to contain this mass virtual force. The major obstacle being how, when these offences are prosecuted, the personal jurisdiction is to be applied.

This article breaks down how the legal principles have evolved while determining personal jurisdiction in cyberspace.

Cyberspace- The Virtual Universe

Cyberspace is an imaginary area or a virtual space where a connection can be established between two computers at any two points in the world, with absolutely no limits.

The word 'cyberspace' was used in the Novel '**Neuromancer**' by **William Gibson**, for the first time in 1984, which is a science fiction and defined as an interaction between the human mind and computers.

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While cyberspace and the internet share very similar connotations, cyberspace can be defined as anything that is done using the internet, while the internet is a network or networks.

In layman terms "cyberspace" is a virtual universe made up of the widely spread and interconnected digital gadgets and technology, enabling one to create, modify, share, exchange, extract and destroy the physical resources floating all over the internet.

The world we live in is possibly at its simplest, most sophisticated version, as at this point in time, and we could only hope for it to make many innovative new changes. The world seems so much smaller at our fingertips, lives have collectively become easier. Education, E-commerce, shopping, banking, and almost every other essential has taken its spot on the internet. In fact, some of the richest multinational companies are that of Google and Facebook that are empires built virtually on nothing but data. The huge number of users are the customers and their personal information, the asset. Each of these businesses run on nothing but loads of information, some private, some not, and it becomes necessary to build a hyper-vigilant screening process in providing our personal information, because of the immense threats that tag-along with this mighty tool.

With business transactions moving online, the conventional methods of dealing with legal complications are also in need of remoulding to fit into the present, needful circumstances.

It is often very ambiguous to decipher what place holds jurisdiction over disputes that arise in the vast cyberspace. In her paper “Principles of Jurisdiction”, Betsy Rosenblatt states that “a court must first decide “where” the internet conduct takes place, and what it means for internet activity to have an “effect” within a state or a nation”.

The concept of national borders and distance stands irrelevant in cyberspace. By setting up a website from a home computer, here in India, one can grant access to anybody around the world, making communication a piece of cake. While communication is easier, the legal threats posed are quite drastic.

Threats To Cyberspace

With the amount of information being constantly exchanged, the threats in cyberspace are equally large. It is also important to register the intensity of changes the cyberspace is constantly subjected to, which concurrently aids in the advancement of the cyberattacks.

Cyberattacks can range from personal data breaches to mass frauds, each of which is equally dangerous and harmful, putting one's usage of cyberspace at risk.

Cyberattacks are where internet users use malicious manoeuvres to steal, destroy, expose, or gain unauthorized access into the personal information of a person, company, military databases, etc.,

Cyberattacks are a part of cyber warfare- where cyberspaces containing classified military information, are attacked to wage war and other military purposes, and cyber terrorism- where cyberspaces are used to conduct violent criminal activities.

Some of these common cyberattacks include phishing, identity theft, ransomware, hacking, child pornography, malware, credit or debit card frauds, disinformation- harming an individual, property or a nation.

Hierarchy of courts, Civil and criminal jurisdictions.

Senior Civil Judge Court, Principal Junior Civil Judge Court and Junior Civil Judge Court are the Subordinate Courts in civil cases. Chief Judicial Magistrate, First Class Judicial Magistrate Court and Second Class Judicial Magistrate Court are the Subordinate Courts in criminal cases.

Jurisdiction of Subordinate Court

The Code of Criminal Procedure provided provisions for the jurisdiction in criminal matters.

[Section 14](#) of the CrPC deals with the local jurisdiction of Judicial Magistrates. This section empowers the Chief Judicial Magistrate, who is subjected to the control of the High Court that he can define the local limits of the areas from time to time, within which the Magistrates exercise all or any of the powers with which they are invested under this code:

1. It is provided that the Special Judicial Magistrate Court may hold its sitting at any place within its local jurisdiction.
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2. If the exception is provided by such definition then the powers of the Magistrate and its local jurisdiction shall extend throughout the district.
3. Where the local jurisdiction of a Magistrate has been extended beyond the district of its jurisdiction or the metropolitan area, as the case may be in which he generally holds court, any reference in this code to the Court of Session, Chief Metropolitan Magistrate or the Chief Judicial Magistrate, in relation to such magistrate, throughout the area which comes under his local jurisdiction, be interpreted, unless the circumstances otherwise requires, as a reference to the Court of Session, Chief Judicial Magistrate, or Chief Metropolitan Magistrate, as the case may be exercising jurisdiction in relation to that district or metropolitan area.

[Section 22](#) of the CrPC deals with the local jurisdiction of Executive Magistrates. This section empowered the District Court, which is subjected to the control of the State Government, that it can draw the local limits of the areas under which the Executive Magistrates may use all or any of the powers with which they may be endowed under this code but there are exceptions when the powers and jurisdiction of such Magistrate shall extend throughout the district.

[Section 27](#) of the CrPC deals with the jurisdiction in the case of juveniles. If the accused is under the age of sixteen years then the case is tried by the Court of the Chief Judicial Magistrate or by any court which is tried under the [Children Act, 1960](#).

[Section 177](#) to [Section 189](#) of the CrPC deals with the provisions related to inquiries and trials of the jurisdiction of the Criminal Courts.

Section 177 of the CrPC provides that the court which comes under the local jurisdiction where the offence has been committed then that offence must be inquired and tried by that court.

[Section 178](#) of the CrPC deals with the provisions related to the place where trial or enquiry of offence should be commenced when there is uncertainty regarding the place of commencement of offence.

[Section 179](#) of the CrPC provides that the trial of the offence is commenced at the place of the act where it is done or the place where the consequence ensues.

[Section 180](#) of the CrPC provided the provisions for a place of trial in a situation where an act becomes offence due to another offence.

In case of certain offences, [Section 181](#) of the CrPC provides provisions for the place of trial for such offences.

[Section 182](#) of the CrPC deals with the offences which are committed by telecommunication messages or by letters etc.

[Section 183](#) of the CrPC deals with the offences which are committed during journey or voyage.

[Section 184](#) of the CrPC deals with the offences which are triable together and provide provisions for such offences.

[Section 185](#) of the CrPC empowered the State Government to direct any cases or class of cases can be tried in a Sessions Court for which the trial has been committed in any district.

[Section 186](#) of the CrPC empowered the High Court to decide the district where the trial or inquiry of offence should be commenced in cases where there is confusion regarding the place of trial.

[Section 187](#) of the CrPC empowers the Magistrate to issue warrant or summons for the offence which is committed beyond the local jurisdiction.

[Section 188](#) of the CrPC describes the offences which are committed outside the territory of India.

[Section 189](#) of the CrPC provides the authority to the Central Government that it can take the receipt of evidence for the offences which are committed outside the territory of India.

The [Code of Civil Procedure, 1908](#), provided provisions for the jurisdiction in case of civil matters.

[Section 15](#) of the CPC provides that the suit for the offence firstly have to be instituted in the Court of the lowest grade competent for the trial.

[Section 16](#) of the CPC provided that where suits have to be instituted, should be based on the subject matter which is subject to the pecuniary or other limitations prescribed by the law.

[Section 17](#) of the CPC provided that the suits for the immovable property have to be filed within the local limits of whose jurisdiction where any part of the property is situated.

[Section 18](#) of the CPC provided provisions for the place of institution of the suit where local limits of the jurisdiction of Courts are uncertain.

[Section 20](#) of the CPC provided provisions for the place of institution of other suits. It states that suits for the offence have to be instituted where the cause of action arises or at the place where the defendants reside.